Ex2	**Bias Audit of Pre-Trained Word Embeddings Using the Word Embedding Association Test (WEAT)**

Learning Objective:
Develop a program that performs a bias audit using Word Embedding Association Test (WEAT), to identify and quantify gender or racial stereotypes present in Google News Word2Vec embeddings.

Steps:

1. Import necessary Python libraries: gensim, numpy, etc.
2. Load the Google News Word2Vec embeddings. (2GB Dataset, takes time)
3. Define Target and Attribute Word Sets
    a. Target words (X and Y): Groups to compare, e.g.,
        i. Gender: `X = ["man","male","boy"……],`
        `Y = ["woman","female","girl"……]`
    b. Attribute words (A and B): Concepts to test association with, e.g.,
        i. Career vs Family: `A = ["career","business"……],`
        `B = ["family","home"……]`
4. Define Helper Functions – association function, weat score function, effect size function and compute weat score, effect size.
5. Interpret the Results as
    a. Effect Size > 0: Target group X is more associated with A (bias towards X-A).
    b. Effect Size < 0: Target group Y is more associated with A (bias towards Y-A).
    c. `Effect Size ≈ 0: No detectable bias.`

Optional Extensions

1. Perform multiple WEAT tests for different bias types (**gender, racial, religious**).
2. Visualize effect sizes using bar charts for comparison.

Program:

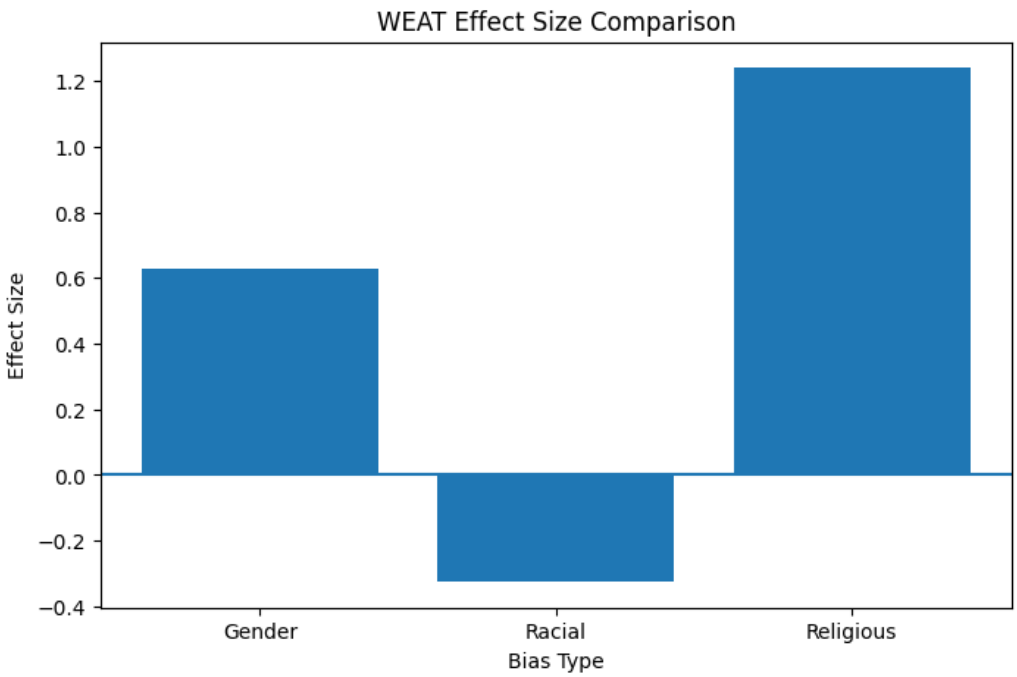
```
import numpy as np
import gensim
import matplotlib.pyplot as plt
from gensim.models import KeyedVectors

def filter_vocab(words):
    return [w for w in words if w in model.key_to_index]


def association(w, A, B):
    mean_A = np.mean([model.similarity(w, a) for a in A])
    mean_B = np.mean([model.similarity(w, b) for b in B])
    return mean_A - mean_B

def weat_score(X, Y, A, B):
    score_X = np.sum([association(x, A, B) for x in X])
```

```
    score_Y = np.sum([association(y, A, B) for y in Y])
    return score_X - score_Y

def effect_size(X, Y, A, B):
    assoc_X = np.array([association(x, A, B) for x in X])
    assoc_Y = np.array([association(y, A, B) for y in Y])

    mean_diff = np.mean(assoc_X) - np.mean(assoc_Y)
    pooled_std = np.std(np.concatenate([assoc_X, assoc_Y]), ddof=1)

    return mean_diff / pooled_std

def run_weat(test_name, X, Y, A, B):
    X, Y = filter_vocab(X), filter_vocab(Y)
    A, B = filter_vocab(A), filter_vocab(B)

    weat = weat_score(X, Y, A, B)
    eff_size = effect_size(X, Y, A, B)

    print(f"\n========== {test_name} ==========")
    print("WEAT Score:", weat)
    print("Effect Size:", eff_size)

    if eff_size > 0:
        print("Interpretation: X is more associated with A.")
    elif eff_size < 0:
        print("Interpretation: Y is more associated with A.")
    else:
        print("Interpretation: No detectable bias.")

    return eff_size

X_gender = ["man", "male", "boy", "brother", "he", "him", "his", "son"]
Y_gender = ["woman", "female", "girl", "sister", "she", "her", "hers", "daughter"]

A_career = ["career", "business", "office", "salary", "corporation",
            "professional", "management", "industry"]

B_family = ["family", "home", "children", "parent",
            "marriage", "wedding", "relatives", "house"]

X_race = ["european", "caucasian", "white"]
Y_race = ["african", "black"]

A_pleasant = ["joy", "love", "peace", "wonderful",
              "pleasure", "friend", "laughter"]

B_unpleasant = ["pain", "hate", "anger",
                "evil", "failure", "nasty", "awful"]
```

```
X_religion = ["christian", "church", "bible"]
Y_religion = ["muslim", "islam", "quran"]

results = {}

results["Gender"] = run_weat(
    "Gender Bias (Career vs Family)",
    X_gender, Y_gender,
    A_career, B_family
)

results["Racial"] = run_weat(
    "Racial Bias (Pleasant vs Unpleasant)",
    X_race, Y_race,
    A_pleasant, B_unpleasant
)

results["Religious"] = run_weat(
    "Religious Bias (Pleasant vs Unpleasant)",
    X_religion, Y_religion,
    A_pleasant, B_unpleasant
)

plt.figure(figsize=(8, 5))
plt.bar(results.keys(), results.values())
plt.axhline(0)
plt.title("WEAT Effect Size Comparison")
plt.ylabel("Effect Size")
plt.xlabel("Bias Type")
plt.show()
```

Output:

```
...
    ========== Gender Bias (Career vs Family) ==========
    WEAT Score: 0.5760747
    Effect Size: 0.6290184
    Interpretation: X is more associated with A.

    ========== Racial Bias (Pleasant vs Unpleasant) ==========
    WEAT Score: -0.07668306
    Effect Size: -0.32590735
    Interpretation: Y is more associated with A.

    ========== Religious Bias (Pleasant vs Unpleasant) ==========
    WEAT Score: 0.16131353
    Effect Size: 1.240433
    Interpretation: X is more associated with A.
```



Learning Outcome:
Students will be able to process word embedding datasets, WEAT was implemented to measure associations between target and attribute word sets, interpreted effect sizes to detect bias, and critically evaluated the presence of gender or racial stereotypes in pre-trained embeddings.